

Hands on Activity. <6.1>

<Function>

Course Code: CPE007

Program: Computer Engineering

Course Title: Programming logic and design

Date Performed: 10/21/2025

Section: CPE11S1

Date Submitted: 10/21/25

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6. Output

[*] funcc.cpp

```

1  #include <iostream>
2  #include <iomanip>
3  using namespace std;
4
5  // Function Prototype
6  void greetUser ();
7  void add(int a, int b);
8  void subtract(int a, int b);
9  void multiply(int a, int b);
10 void divide(int a, int b);
11 void fahrenheitToCelsius(float f);
12 void celsiusToFahrenheit(float c);
13 void dollarsToPesos(float dollars);
14 void pesosToDollars(float pesos);
15
16 // Main Function
17 int main () {
18     int choice;
19     int num1, num2;
20     float temp, amount;
21
22     cout << fixed << setprecision(2);
23
24     do {
25         cout << "===== FUNCTION MENU =====" << endl;
26         cout << "1. Add, Subtract, Multiply, or Divide Two Integers. " << endl;
27         cout << "2. Convert Fahrenheit to Celsius and Celsius to Fahrenheit. " << endl;
28         cout << "3. Convert Dollar to Peso and Peso to Dollar. " << endl;
29         cout << "4. Exit" << endl;
30         cout << "Enter your choice (1-4): ";
31         cin >> choice;
32
33         switch(choice) {
34             case 1:
35                 cout << "Enter two integers: ";
36                 cin >> num1 >> num2;
37                 cout << "Choose operation (1. Add, 2. Subtract, 3. Multiply, 4. Divide): ";
38                 int operation;
39                 cin >> operation;
40                 switch (operation) {
41                     case 1: add(num1, num2); break;
42                     case 2: subtract(num1, num2); break;
43                     case 3: multiply(num1, num2); break;
44                     case 4: divide(num1, num2); break;
45                     default: cout << "Invalid operation!" << endl; break;
46                 }
47                 break;

```

[*]funcc.cpp

```
49     case 2:
50         cout << "Enter temperature in Fahrenheit: ";
51         cin >> temp;
52         fahrenheitToCelsius(temp);
53         cout << "Enter temperature in Celsius: ";
54         cin >> temp;
55         celsiusToFahrenheit(temp);
56         break;
57
58     case 3:
59         cout << "Enter amount in dollars: ";
60         cin >> amount;
61         dollarsToPesos(amount);
62         cout << "Enter amount in pesos: ";
63         cin >> amount;
64         pesosToDollars(amount);
65         break;
66
67     case 4:
68         cout << "Exiting program. Goodbye!" << endl;
69         break;
70
71     default:
72         cout << "Invalid choice! Please enter a number between 1 and 4." << endl;
73         break;
74 }
75 } while (choice != 4);
76
77 return 0;
78 }
79
80 // Function Definitions
81 void add(int a, int b) {
82     cout << "Sum: " << (a + b) << endl;
83 }
84
85 void subtract(int a, int b) {
86     cout << "Difference: " << (a - b) << endl;
87 }
88
89 void multiply(int a, int b) {
90     cout << "Multiplied: " << (a * b) << endl;
91 }
92
93 void divide(int a, int b) {
94     if (b != 0) {
95         cout << "Divided: " << (float)a / b << endl;
```

```
96 |     } else {
97 |         cout << "Error: Division by zero!" << endl;
98 |     }
99 | }
100 |
101 | void fahrenheitToCelsius(float f) {
102 |     float c = (f - 32) * 5 / 9;
103 |     cout << "Fahrenheit " << f << " is " << c << " Celsius." << endl;
104 | }
105 |
106 | void celsiusToFahrenheit(float c) {
107 |     float f = (c * 9 / 5) + 32;
108 |     cout << "Celsius " << c << " is " << f << " Fahrenheit." << endl;
109 | }
110 |
111 | void dollarsToPesos(float dollars) {
112 |     float rate = 18.50;
113 |     cout << "$" << dollars << " is " << dollars * rate << " pesos." << endl;
114 | }
115 |
116 | void pesosToDollars(float pesos) {
117 |     float rate = 18.50;
118 |     cout << pesos << " pesos is $" << pesos / rate << endl;
119 | }
```

```

C:\Users\TIPQC\Downloads\S: X + v
===== FUNCTION MENU =====
1. Add, Subtract, Multiply, or Divide Two Integers.
2. Convert Fahrenheit to Celsius and Celsius to Fahrenheit.
3. Convert Dollar to Peso and Peso to Dollar.
4. Exit
Enter your choice (1-4): 1
Enter two integers: 21
12
Choose operation (1. Add, 2. Subtract, 3. Multiply, 4. Divide): 3
Multiplied: 252
===== FUNCTION MENU =====
1. Add, Subtract, Multiply, or Divide Two Integers.
2. Convert Fahrenheit to Celsius and Celsius to Fahrenheit.
3. Convert Dollar to Peso and Peso to Dollar.
4. Exit
Enter your choice (1-4): 2
Enter temperature in Fahrenheit: 32
Fahrenheit 32.00 is 0.00 Celsius.
Enter temperature in Celsius: 43
Celsius 43.00 is 109.40 Fahrenheit.
===== FUNCTION MENU =====
1. Add, Subtract, Multiply, or Divide Two Integers.
2. Convert Fahrenheit to Celsius and Celsius to Fahrenheit.
3. Convert Dollar to Peso and Peso to Dollar.
4. Exit
Enter your choice (1-4): 3
Enter amount in dollars: 3445
$3445.00 is 63732.50 pesos.
Enter amount in pesos: 32321
32321.00 pesos is $1747.08
===== FUNCTION MENU =====
1. Add, Subtract, Multiply, or Divide Two Integers.
2. Convert Fahrenheit to Celsius and Celsius to Fahrenheit.
3. Convert Dollar to Peso and Peso to Dollar.
4. Exit
Enter your choice (1-4): |

```

7. Supplementary Activity

I created a void function, a prototype function which my function menu, like the add, subtract, divide, multiply, etc. Then in my main function first put the an int to identify my choice so I can put it in **cin** then again I use int to identify the num 1 and num 2, then float for me to have a decimal output. Then using my **iomanip** I set the `<<fixed<<setprecision`, I forgot what is called for that function but I remember how it works, then in my **cout** I use a **do** to initiate my **cout** then I put my function inside **cout**,.... then in my **switch**, I use my first int which is the choice, then case 1 its just a simple output, **cout** for the print then **cin** for the user so the user can type out,then another **cout** so the user can choose what operation he/she wants. Then I create an int operation because I will need it to so the user can choose what operation, then the **cin**, that's when the user can put the operation, then **switch** operation, automatic if they put two integer it will automatically break or show the answer, it will show what the operation you choose, then for the default it well print out Invalid operations if you type some errors or just typo, then **break** so we can go to case 2. this in the case 2 it will calculate the fahrenheit

and the celcius but the output is basic like the first case, you will just type the fahrenheit and celsius, move to the next case 3, it's the same in case 2, and the case 4 if you choose the number 4 it will automatically exit, then in the default if you choose 6 or other tan 1-4 it will print out the invalid choice, then in thw while, it will keep running until the user choose exit which is the number 4, then we create a function definitions so it will add, subtract etc, first I put the sum function, void add (+) like that to sum up. It's the same code in the add, subtract, multiply you just need to change the symbols, but in the division I put a float and if because there is a chance that it will have a decimal, then else to print out if its Error, then, the function of fahrenheit to celcius I created a float f that will assume the rate, then if the user put the number it will automatically change the number and it will print what user inputs, Then I repeat it in the second one the celcius to fahrenheit. Then in the dollar to peso it will calculate it and I do the same in the last code, but in the dollar to peso its multiply and in peso to dollar I use divide, then end of the code.

8. Conclusion

In my conclusion, I learned about void, it makes your code much easier and shorter, because if I don't use void in here, it will look I code too much, then learned about the prototypes but the topic I understand in this code very well is the usage of the function definition, but overall I learned here and I gain knowledge.

9. Assessment Rubric